

LOT-HW-20261009-01

LOT-HW-20261009-01 — LOT Computer (hardware) · Session 01 Report

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Date: 2026-10-09 · **Branch:** claude/brave-lamport-mctw4d **Status:** PLAN + BOM v0.1 + specs. Nothing is ordered, fabricated or tested.

0. Honesty box (read first)

- lot-systems.com, brand.lot-systems.com, institute.lot-systems.com do **not resolve from this sandbox**. Brand/about/CQGS pages were NOT read. Brand cues come from the repo (docs/technical/LOT-STYLE-GUIDE.md, LOT-NODE-0-RIG-SPEC.md). Re-check against brand.lot-systems.com before any engraving/logo work.
- Prices in BOM-v0.1.csv are **order-of-magnitude estimates from memory, not quotes**. Vendor links are search/landing pages, not verified listings. Treat as “where to look”, get real quotes in Phase 1.
- Total BOM estimate: **~\$96/unit, ~\$9.6k for 100 units** (parts only, excl. NRE, tooling, shipping, duties, yield loss). Realistic all-in for 100 units: **\$15–25k incl. NRE and 10–20% scrap**.

1. Product definition

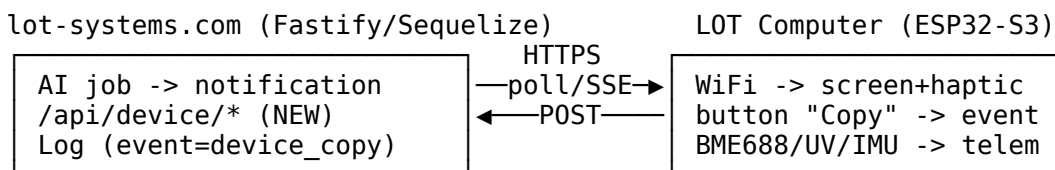
LOT Computer — a 40 × 40 mm pocket “pager” made of two polished stainless-steel parts. Face A: camera, small screen, one button. Face B: polished mirror steel. It receives autonomous AI notifications from lot-systems.com (e.g. “Coffee time!”), senses weather/environment, and its button (“Copy”) posts a signal back to the **Log tab**. Wireless-charged.

2. Requirement traceability (your 19 points)

#	Requirement	Where addressed	Status
1	PCBWay	PCB + SMT assembly + CNC (BOM PCB1/PCB2/BODY)	Plan
2	Pager-like notification from AI site	API-CONNECTOR-SPEC §notifications, M1/BZ1	Spec
3	2-part stainless body	§4 mechanical	Plan, risk
4	Flat silver square 4×4 cm × 5 mm	§4 — 5 mm is not achievable for v1	Risk
5	Camera	CAM1 OV2640	Plan
6	LOT API connector	API-CONNECTOR-SPEC	Spec
7	PDF manuals	pdf/ build (this session)	Done (draft)
8	Compress info each session	SESSION-STATE.md	Done
9	Firmware documents	FIRMWARE-SPEC	Done (draft)

#	Requirement	Where addressed	Status
10	Software to connect with firmware	API spec §companion + FW §host tool	Spec
11	Separate documents	4 separate .md + PDFs	Done
12	Charger	CH1 + U6	Plan
13	100-unit run	§6	Plan
14	Weather sensor	U2 BME688 (+ site weather API)	Plan
15	AI-grade off-the-shelf sensors	BME688 (BSEC AI), BMI270, LTR-390	Plan
16	“Copy” button -> Log tab	API spec device_copy event	Spec; needs server change
17	One side polished stainless	BODY2	Plan
18	Other side camera+screen+button	BODY1 + G1	Plan
19	Wireless charger	U7/L1/CH1	Plan, RF risk

3. Architecture



Existing repo facts used: Log model has `userId`, `text`, `event`, `metadata(JSONB)`, `context`; `/api/logs` only returns events in a `displayableEvents` whitelist (`src/server/routes/api.ts:1082`); `/api/live-message` already exists (GET, text only); all `/api` routes are session-cookie authenticated (`req.user`), so **devices need a new token auth path** — not present today.

4. Mechanical & RF — the hard truth

- 5 mm total thickness is not realistic for v1.** Stack: cover glass 0.5 + display ~1.5–2 + camera module ~3–4 + PCB 0.8 + battery ≥2 + coil/ferrite ~1 + two steel skins ≥0.6 each. Realistic v1: **8–10 mm**. A 5 mm unit needs flex-PCB, bare-die camera, no haptic motor, ~100 mAh cell. Recommendation: build v1 at ~9 mm, and treat 5 mm as v2 with a separate feasibility study.
- Stainless steel blocks WiFi/BLE and Qi.** A fully closed metal shell kills both. Solution: a non-metal window (glass/zirconia/PEEK, WIN1) over the antenna and Qi coil, or a slotted/segmented steel edge. Polished side must stay metal (req 17) → put coil+antenna behind the cover-glass face (face A) and charge **screen-down** on the dock, or use a glass insert ring on the mirror side. Prototype both before tooling.
- Use 316L, mirror-polished ($Ra < 0.05 \mu m$). 2 parts: bezel/front + back, laser-welded or screwed (4 × M1.0 hidden under bezel). IP rating: target IPX4 v1.
- Camera behind cover glass with small aperture; check privacy — add physical shutter or LED indicator (recommended).

5. Roadmap (analysis)

Phase	Window	Output	Gate
0 Spec	Oct 2026	This doc, BOM v0.1, API spec	Vadik approves
1 Quotes & dev kit	Oct–Nov	PCBWay CNC/PCB quotes; ESP32-S3 + BME688 + display breadboard	Real prices, RF test through glass/steel
2 Server API	Nov	/api/device/*, token auth, device_copy log event, tests	Benchmark green
3 Firmware alpha	Nov–Dec	Wi-Fi provisioning, notify, button, sensors	Notification round trip < 5 s
4 EVT (10 units)	Dec–Jan	Proto PCB + 3D-printed/CNC body	Qi + WiFi pass in enclosure
5 DVT (25 units)	Feb–Mar 2027	Final steel, polish, drop/IP tests	Fix list closed
6 PVT / 100-unit run	Apr–Jun 2027	100 assembled, PDF manuals, boxes	Yield $\geq$ 85%

Critical path: steel RF design (Phase 1) and CNC polished-part lead time (3–5 weeks).

6. 100-unit run plan

Order ~120 PCBs/BOM (20% attrition), PCBWay turnkey SMT; CNC/MIM 316L 120 sets; manual polishing is the cost driver (**ask for a quote with polish spec**). Functional test jig (pogo pads) flashes + tests every unit and writes a serial to the site. Components at risk of shortage: BME688, Qi RX IC, thin LiPo — buy 150 pcs up front.

7. Risks

RF/Qi through steel (high) · 5 mm thickness (high) · camera privacy/regulatory FCC/CE/UL for battery + radio (medium; 100-unit run still needs FCC Part 15 declaration — budget \$3–10k) · li-ion shipping rules (medium) · server has no device auth (medium, in our control).

8. Next session (proposed)

1. Get real quotes (needs network access to vendors).
2. Implement /api/device/* + tests behind a feature flag.
3. KiCad schematic skeleton.
4. Confirm with Vadik: is 8–10 mm acceptable for v1? Screen-down charging OK?

9. Deliverables in this session

BOM-v0.1.csv · API-CONNECTOR-SPEC.md · FIRMWARE-SPEC.md · SESSION-STATE.md · pdf/*.pdf